

Mohamed Abdellatif Etman

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PROFESSIONAL SUMMARY

AI Engineer with experience building and deploying production AI systems including Generative AI applications, RAG pipelines, multilingual conversational agents, and multimodal AI solutions. Specialized in deploying scalable AI systems for Arabic language processing, conversational AI, and multimodal automation. Skilled in FastAPI, Azure Cloud, vector search, and LLM integration with practical experience delivering scalable AI applications across enterprise, finance, and government projects. Background in full-stack development using Python, React Native, and Odoo.

EDUCATION

Diploma in AI, Data Science, and Cloud Computing Jan 2024 – Dec 2024
Digital Egypt Builders Initiative (DEBI), Ministry of Communications Hybrid, Egypt

- Intensive 12-month program covering Machine Learning, Deep Learning, NLP, Computer Vision, Cloud Computing, and MLOps.

Bachelor’s Degree in Computer Science Jul 2019 – Jul 2023
Faculty of Computers and Information, Tanta University Tanta, Egypt

- Relevant Coursework: Algorithms, Data Structures, Database Systems, Software Engineering, Artificial Intelligence, and Machine Learning.

WORK EXPERIENCE

AI Engineer Oct 2025 – Present
Integra Pro Solutions Cairo, Egypt

- Design and develop AI-powered business automation features integrated with Odoo ERP platforms.
- Build full-stack applications using Python, Django, React Native, and Odoo with custom API integrations.
- Implement Generative AI workflows for intelligent document processing, analytics, and operational automation.

AI Engineer (Freelance) Dec 2024 – Oct 2025
Independent Contractor Remote

- Designed and deployed production AI applications including multilingual chatbots, RAG systems, and voice-enabled assistants for tourism and enterprise clients.
- Built scalable backend services using FastAPI, Redis, and asynchronous processing for real-time AI interactions.
- Implemented webhook-driven messaging workflows using Meta Graph API and WhatsApp Business API for omnichannel conversational AI.
- Fine-tuned and evaluated open-source LLMs including Llama 3 and Qwen for Arabic conversational AI workflows.

AI Developer – MCIT Innovation Project Nov 2024 – Mar 2025
Ministry of Communications and Information Technology (MCIT) Remote, Egypt

- Contributed to the development of a multimodal AI system for automated employee performance analysis using video, audio, and text data.
- Built ML pipelines covering preprocessing, model evaluation, and Azure-based deployment workflows.
- Applied multimodal fusion and inference optimization techniques, reducing processing time by 40%.
- Benchmarked LLMs including Gemini, Llama, and Qwen to optimize inference quality and deployment cost.

GenAI Engineer (Internship) Jul 2024 – Oct 2024
National Bank of Egypt (NBE) Remote, Egypt

- Developed an AI-powered financial dashboard analyzer using Vision Transformers and NLP techniques for chart interpretation.
- Designed hybrid AI pipelines combining computer vision and retrieval-based NLP for automated financial insights.
- Integrated Azure Cognitive Services for bilingual voice-enabled interactions in Arabic and English.

– Reduced financial analyst review time by 60% through workflow automation and intelligent report generation.

Mobile App Developer & Team Lead

Aug 2022 – Sep 2025

Camp Coding

Tanta, Egypt

- Developed and maintained cross-platform mobile applications using React Native.
- Led small development teams and coordinated feature delivery using Agile methodologies.
- Integrated REST APIs, push notifications, media streaming, and real-time features into production applications.

KEY PROJECTS

EduVision – Multimodal AI Performance Evaluation System

MCIT Government Project

- Built a multimodal AI pipeline analyzing video, audio, and text data to automate employee performance evaluation workflows.
- Integrated OpenPose, MediaPipe, Whisper, and fine-tuned Llama 3 / Qwen-VL models for behavioral and speech analysis.
- Optimized Azure deployment workflows and reduced processing time by 40% through inference optimization techniques.
- Benchmarked Gemini, Llama, and Qwen models to balance inference quality, latency, and deployment cost.
- **Technologies:** Python, PyTorch, OpenPose, MediaPipe, Whisper, Llama 3, Qwen-VL, FastAPI, Azure Cloud, Docker.

Smart Tourism Assistant – RAG-Powered Omnichannel Chatbot

- Designed and deployed a multilingual conversational AI assistant supporting WhatsApp, Messenger, Instagram, and Web platforms.
- Implemented hybrid RAG pipelines using ChromaDB semantic retrieval and HuggingFace reranking models for Arabic query optimization.
- Built FastAPI backend services with Redis session persistence and Meta Graph API webhook integrations.
- Automated customer support workflows and reduced response times across tourism service channels.
- **Technologies:** Gemini, GPT-4o, FastAPI, Redis, ChromaDB, LangChain, Docker, Meta Graph API, WhatsApp Business API.

Financial Dashboard Auto Transcript Analyzer

National Bank of Egypt

- Developed a GenAI-powered analytics platform for extracting and interpreting financial chart data from dashboard images.
- Implemented Vision Transformer models (Donut) for chart understanding and structured data extraction from 500+ financial chart images.
- Built retrieval-based Q&A workflows combining vector search and LLM-based financial analysis.
- Integrated Azure TTS and translation services for bilingual voice interaction and reporting.
- Reduced analyst review time by 60% through automation and intelligent report generation workflows.
- **Technologies:** Transformers, Donut, ChromaDB, GPT-NeoX-20B, FastAPI, Azure Cognitive Services, PyQt5, Python.

TECHNICAL SKILLS

Generative AI & LLMs: GPT-4o, Gemini, Claude, Llama 3, Qwen-VL, RAG, LangChain, Prompt Engineering, LoRA, QLoRA

Machine Learning & Deep Learning: PyTorch, TensorFlow, Scikit-learn, Transformers, CNNs, RNNs, BERT, Transfer Learning

Computer Vision & Multimodal AI: OpenCV, OpenPose, MediaPipe, YOLO, Vision Transformers

Natural Language Processing: Hugging Face, spaCy, NLTK, Sentiment Analysis, Named Entity Recognition

Backend Development: FastAPI, Django, Flask, REST APIs, WebSockets, Redis, Celery

Databases & Vector Search: PostgreSQL, MongoDB, MySQL, ChromaDB, Qdrant, pgvector

Frontend & Mobile: React Native, React.js, Streamlit, PyQt5

Cloud & DevOps: Microsoft Azure, Azure OpenAI, Azure ML, Docker, Git, GitHub, CI/CD, MLflow

Programming Languages: Python, JavaScript, TypeScript, PHP, SQL, Bash

ERP Systems: Odoo (ORM, XML-RPC, JSON-RPC, Custom Modules)

CERTIFICATIONS

Microsoft Azure AI-102: AI Engineer Associate

LANGUAGES

Arabic: Native

English: Professional Working Proficiency